Raw Socket

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Part 1:

Introduction to mininet

Introduction to Raw Socket

What is Socket?

A socket is an interface between an application and the network.

Types:
- TCP Socket
- UDP Socket
- Raw Socket

Traditional Socket Communication:

Application -> TCP/UDP -> IP -> NIC
the operating system automatically creates TCP, UDP, and IP headers.
The application only sends data and does not control packet headers.

What is a Raw Socket?

A raw socket provides direct access to network packets and protocol headers.

Features
- Access to IP headers
- Access to protocol information
- Packet inspection
- Custom packet generation

Raw sockets allow applications to see much more information than ordinary sockets. They expose details usually hidden by the operating system.

Why Do We Need Raw Sockets?

Uses

- Network monitoring
- Protocol analysis
- Packet sniffing
- Security tools
- Research and experimentation

Many networking and security tools depend on raw sockets because they need access to actual packets instead of only application data.

Feature	TCP	UDP	Raw
Reliable	Yes	No	User controlled
Header Access	No	No	Yes
Packet Creation	No	No	Yes
Complexity	Low	Low	High

Raw sockets provide the greatest flexibility but also require deeper networking knowledge.

Position of Raw Sockets

Linux Network stack

Application
|
Socket Layer
|
TCP / UDP
|
IP Layer
|
Data Link Layer
|
Hardware

Raw Socket

Application
|
Raw Socket
|
IP
|
Ethernet
|
NIC

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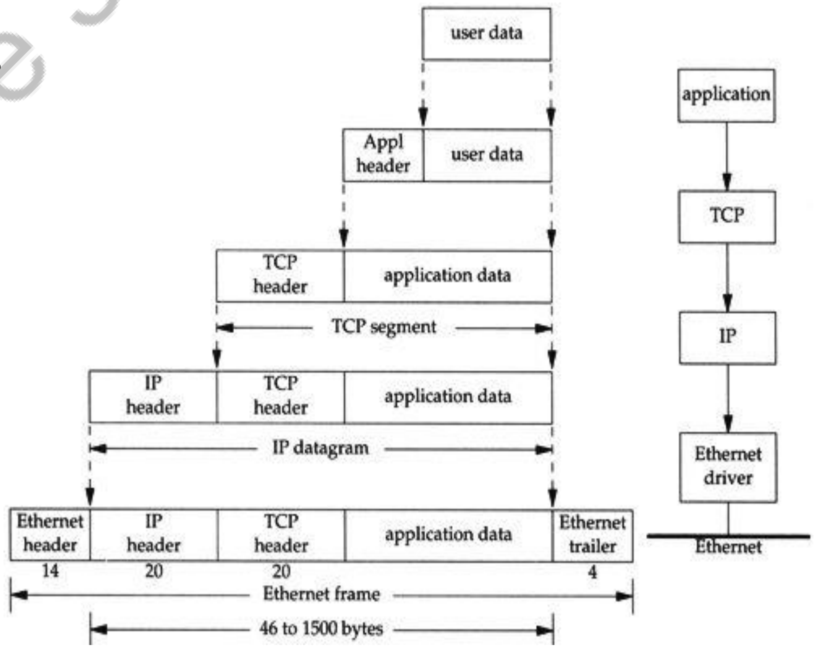
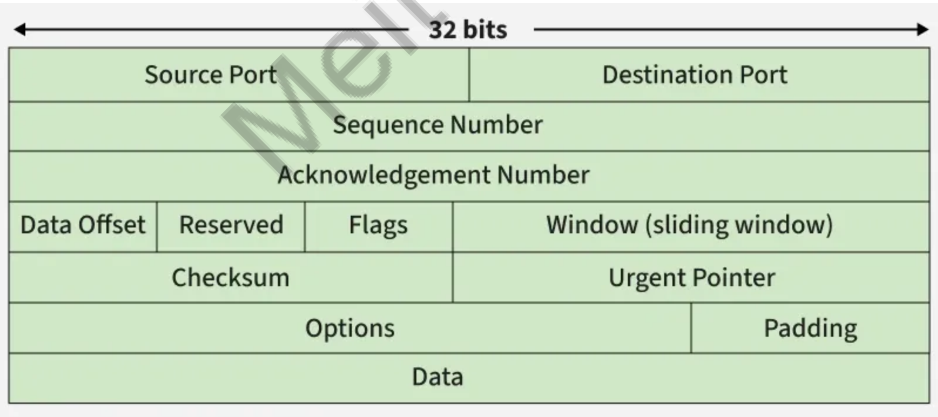
More about Raw Sockets

```
+-------------------+
| Ethernet Header   |
+-------------------+
| IP Header         |
+-------------------+
| TCP Header        |
+-------------------+
| Data              |
+-------------------+
```

Each network packet contains multiple headers.
Raw sockets allow us to inspect these headers.





Raw Socket: Sender - Receiver program

The key difference is the use of SOCK_RAW. The protocol field determines which packets will be captured.

```
int sockfd;

sockfd = socket(
   AF_INET,
   SOCK_RAW,
   IPPROTO_TCP
);
```

You need Root Privileges: why?

Raw sockets can

- Observe packets
- Modify headers
- Create custom packets

Run as:

```
sudo ./program
```

Because raw sockets can potentially be misused, Linux restricts them to privileged users.

Applications

Networking
- Packet analyzers
- Protocol debugging
- Traffic monitoring

Security
- Intrusion Detection Systems
- Traffic analysis
- DDoS detection

Wireshark	Raw Socket Program
Ready-made tool	Custom implementation
GUI	Code based
Automatic decoding	Manual decoding

Many well-known networking and security tools rely on raw packet access.

Advantages and Limitations

Advantages

- Full packet visibility
- Protocol experimentation
- Educational value

Limitations

- Root access required
- More complex
- OS dependent

Raw sockets are powerful but require careful programming and networking knowledge.

Thank You For Joining Us

We hope this lecture inspired curiosity, equipped you with knowledge, and encouraged you to explore further digital adventures excitingly, together.